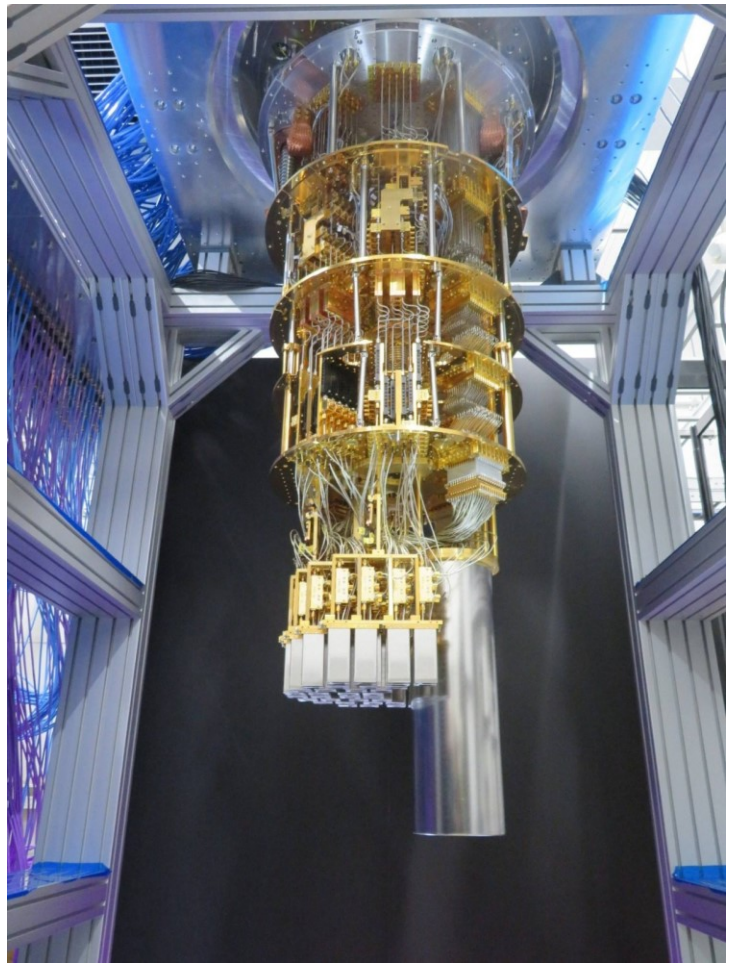
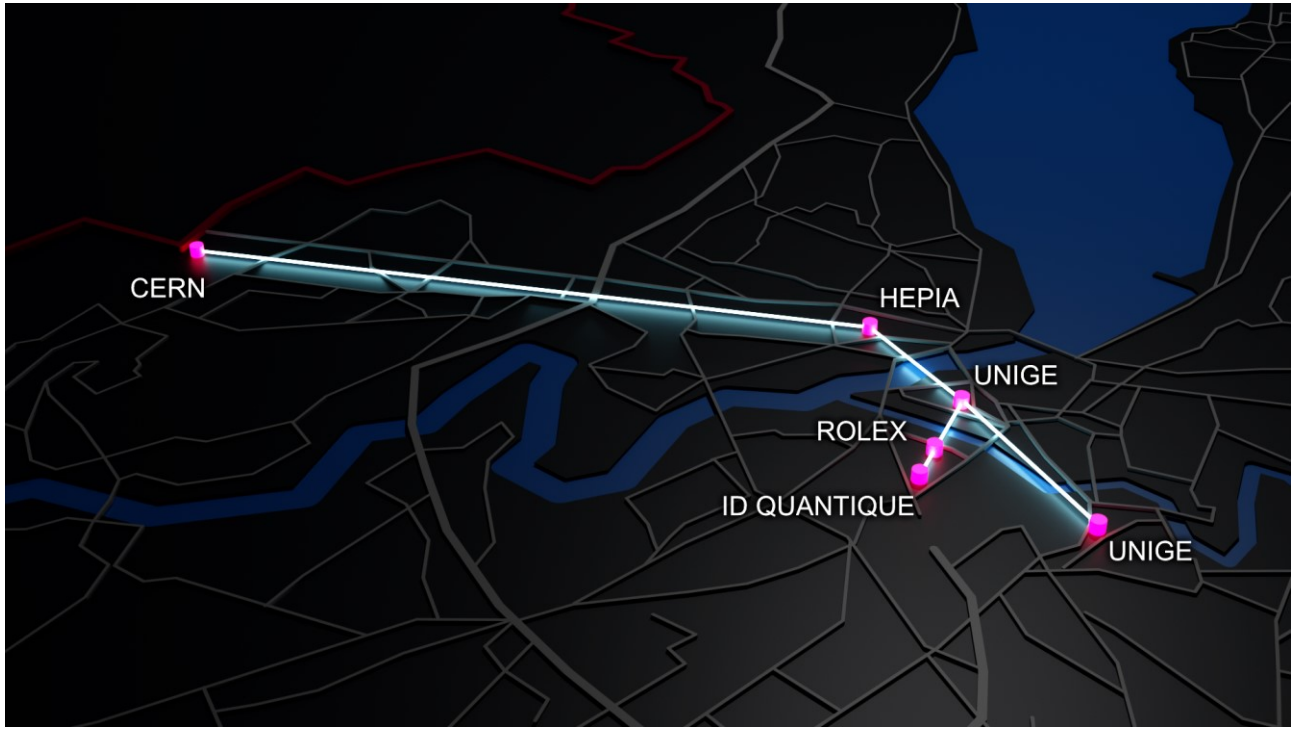
What is Quantum Computing ?

Rafael Tiedra, March 2026



quantum computer from
Fujitsu and Riken Institute
(April 2025)

Latest news !



Geneva Quantum Network (illustration Xavier Ravinet, October 2025)

Nobel Prize in Physics 2025

Ill. Niklas Elmehed © Nobel Prize Outreach
John Clarke
Prize share: 1/3

Ill. Niklas Elmehed © Nobel Prize Outreach
Michel H. Devoret
Prize share: 1/3

Ill. Niklas Elmehed © Nobel Prize Outreach
John M. Martinis
Prize share: 1/3

The Nobel Prize in Physics 2025 was awarded jointly to John Clarke, Michel H. Devoret and John M. Martinis "for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit"

Royal Swedish Academy of Sciences (October 2025)

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Introduction

Quantum computing...

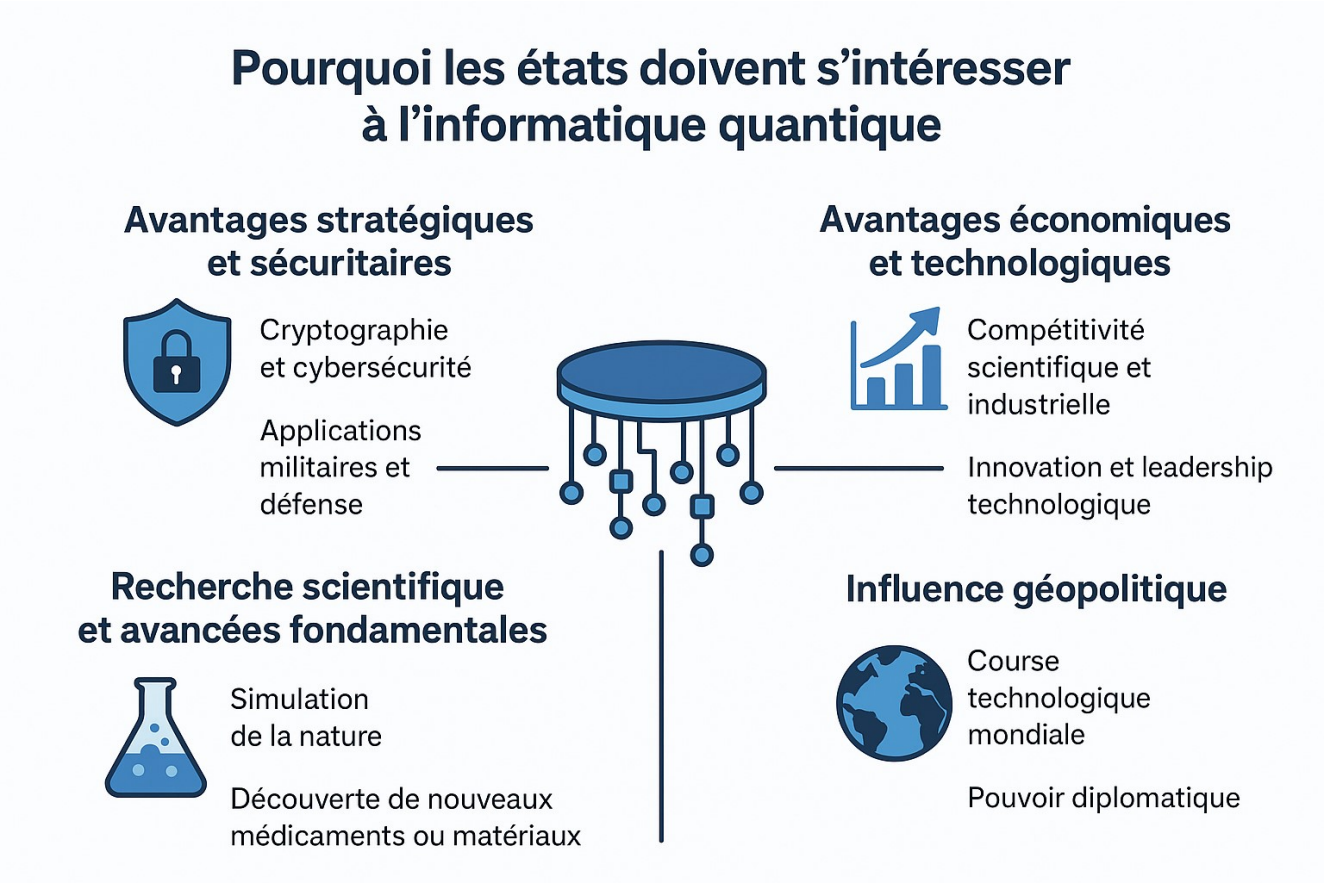
❑ Why do I care ?

Technology watch at OCSIN, 20 years of research in mathematical physics, currently writing a monograph on quantum mechanics with a professor at Nagoya University

❑ Why should organisations care ?

National security agencies (NCSC in the UK, NIST in the US, etc.) estimate that quantum computers could be able of breaking current encryption systems by 2035. But there is much more :

illustration with ChatGPT (in French, September 2025)



Introduction

What is quantum computing ?

Quantum computing is a branch of computer science that relies on the principles of quantum mechanics to encode and process information.

Unlike classical computing, which uses bits (0 or 1), it uses qubits governed by the laws of quantum mechanics.

→ Will make possible to solve <u>some problems</u> beyond the capabilities of classical computers.

Fundamental principles

A qubit («quantum bit») is the unit of information in quantum computing. It can be in the states $|0\rangle$, $|1\rangle$, or in a superposition

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with α and β complex amplitudes, $|\alpha|^2 + |\beta|^2 = 1$.

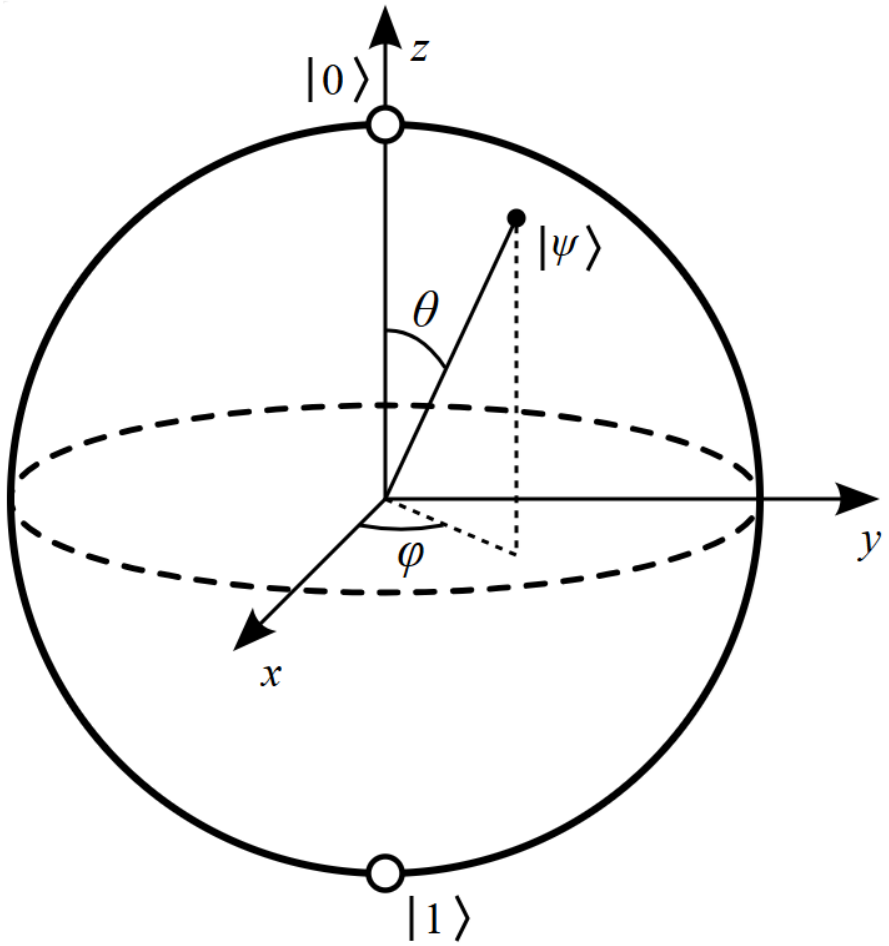
($|\psi\rangle$ can be represented as a point on a sphere of radius 1 called the « Bloch sphere»)

Several approaches to manufacturing qubits :

- ❑ Supraconductor qubits (Google, IBM, Rigetti, Intel)
- ❑ Trapped ions (IonQ, Quantinuum)
- ❑ Spin qubits (Intel, Silicon Quantum Computing)
- ❑ Photonic qubits (Xanadu)

Not yet clear which technologies will prevail…

Bloch sphere (Wikipedia, September 2025)



Fundamental principles

The principles of quantum mechanics used in quantum computing are (among others) :

❑ Superposition and interference of wave functions
(qubits can be in superposition of several states simultaneously)

For example : $\left(\frac{1}{2}|0\rangle + \frac{1}{4}|1\rangle\right) + \frac{1}{4}|0\rangle = \frac{3}{4}|0\rangle + \frac{1}{4}|1\rangle$

❑ Entanglement of wave functions
(qubits can be correlated «instantly», even at a distance)

For example : $\frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$ (famous example called «Bell state»)

❑ Collapse of wave functions during a measurement
(the classical state of a quantum system is determined when it is measured/observed 🐈)

For example : $\frac{1}{2}|0\rangle + \sqrt{\frac{3}{4}}|1\rangle \longrightarrow$ $\longrightarrow |1\rangle$

Fundamental principles

Quantum logic gates («quantum gates») are the quantum equivalent of classical logic gates such as AND, OR, NOT, etc., but they operate on qubits rather than bits.

Quantum gates...

- ❑ transform the state of qubits in a quantum circuit
- ❑ perform reversible transformations (unlike classical gates, which can be irreversible)
- ❑ are represented mathematically by unitary matrices (which guarantees reversibility + conservation of the norm of wave functions)

Fundamental principles

Unitary matrices stem from the Schrödinger equation in finite dimension...

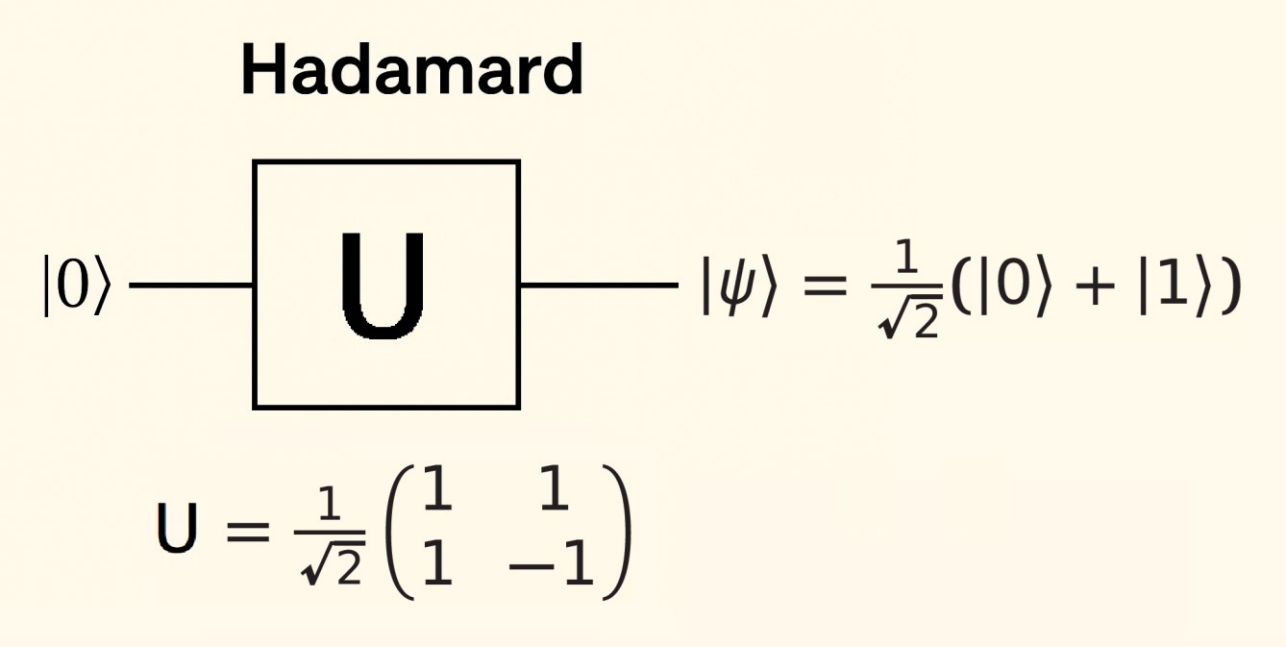
$$i\,\frac{\mathrm{d}}{\mathrm{d}t}\,|\psi(t)\rangle = H\,|\psi(t)\rangle \iff |\psi(t)\rangle = \mathrm{e}^{-itH}|\psi(0)\rangle \quad (H = \text{hermitian matrix})$$

... and in discrete time $t = n$:

$$|\psi(n)\rangle = (\mathrm{e}^{-iH})^n|\psi(0)\rangle \iff |\psi(n)\rangle = U^n|\psi(0)\rangle \quad (U = \text{unitary matrix})$$

Fundamental principles

Example of quantum gate :

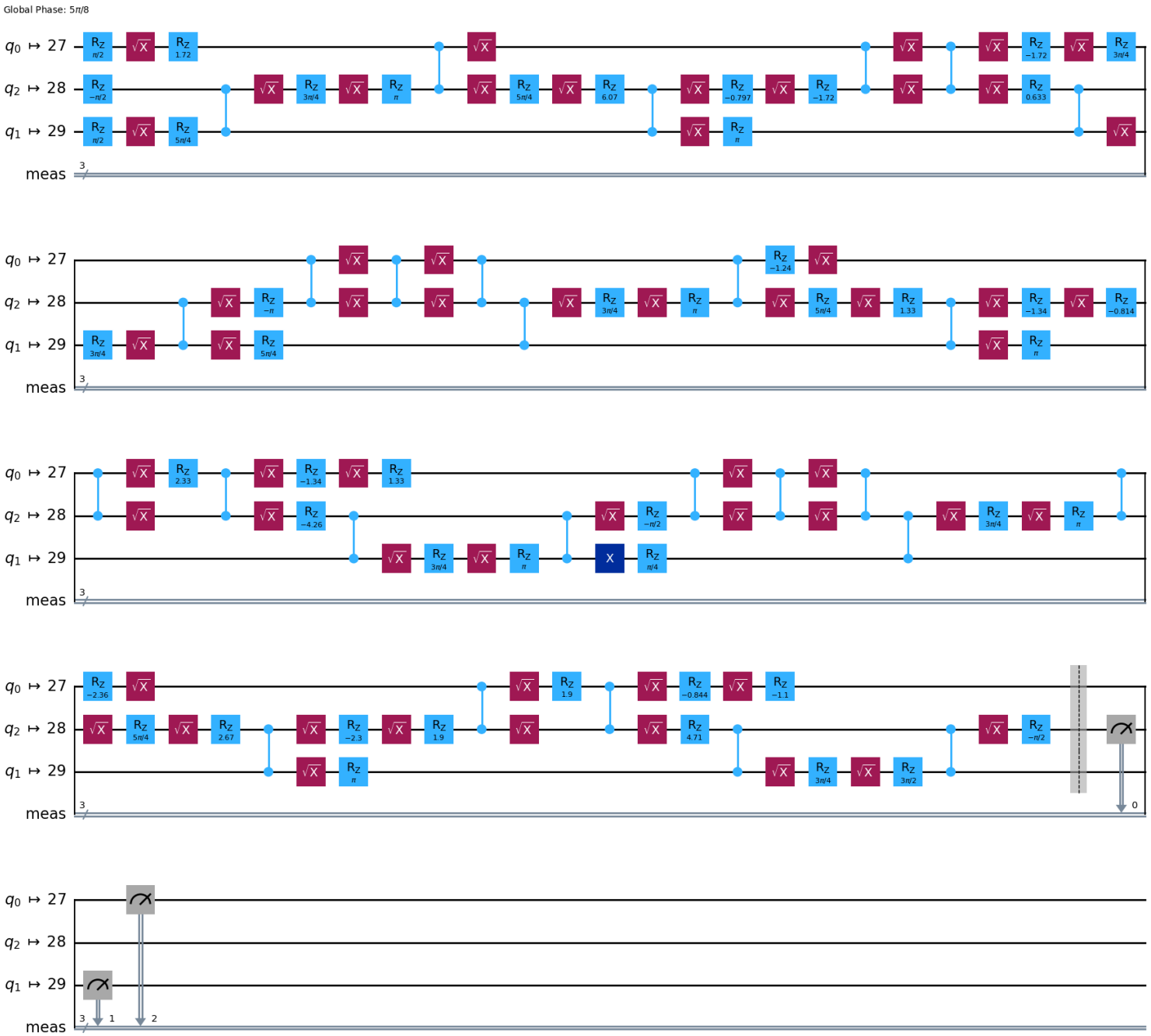


Fundamental principles

Example of a 3-qubit circuit executed on the «ibm_sherbrooke» quantum computer on May 7, 2025.

Programming and diagram created using the open-source Python library Qiskit, introduced by IBM Research in 2017.

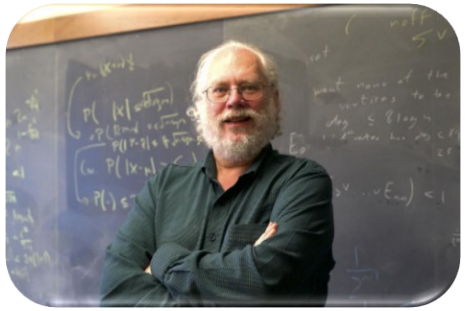
(diagram to be read like a musical score...)



Applications

Two iconic algorithms illustrate the risks and opportunities brought by quantum computing.

CERN (March 2021)



❑ **Shor's algorithm («risk») :**
Makes possible (with a fully functional quantum computer) the factorisation of large numbers in exponentially less time than with classical computers.

→ Will make possible breaking various public key encryption protocols such as RSA, Diffie-Hellman, elliptic curves, etc.

dotquantum.io (unknown date)

❑ **Grover's algorithm («opportunity») :**
Makes possible (with a fully functional quantum computer) the search for an element in an unordered list in quadratically less time than with classical computers.

→ Will improve the speed of multiple industrial applications based on searching for an optimal element/configuration in a large set.

(the previous circuit implements Grover's algorithm for a list of $2^3 = 8$ éléments 😀)

Applications

There are numerous areas of application for quantum computing, at various stages of development (academic research, proof of concept, or already deployed) :

- ❑ Cryptography and cybersecurity
- ❑ Optimisation and operational research
- ❑ Simulation of physical and chemical systems
- ❑ AI and Machine Learning
- ❑ Data science and information retrieval
- ❑ Quantum communication
- ❑ Quantum metrology and sensors
- ❑ Finance and economics

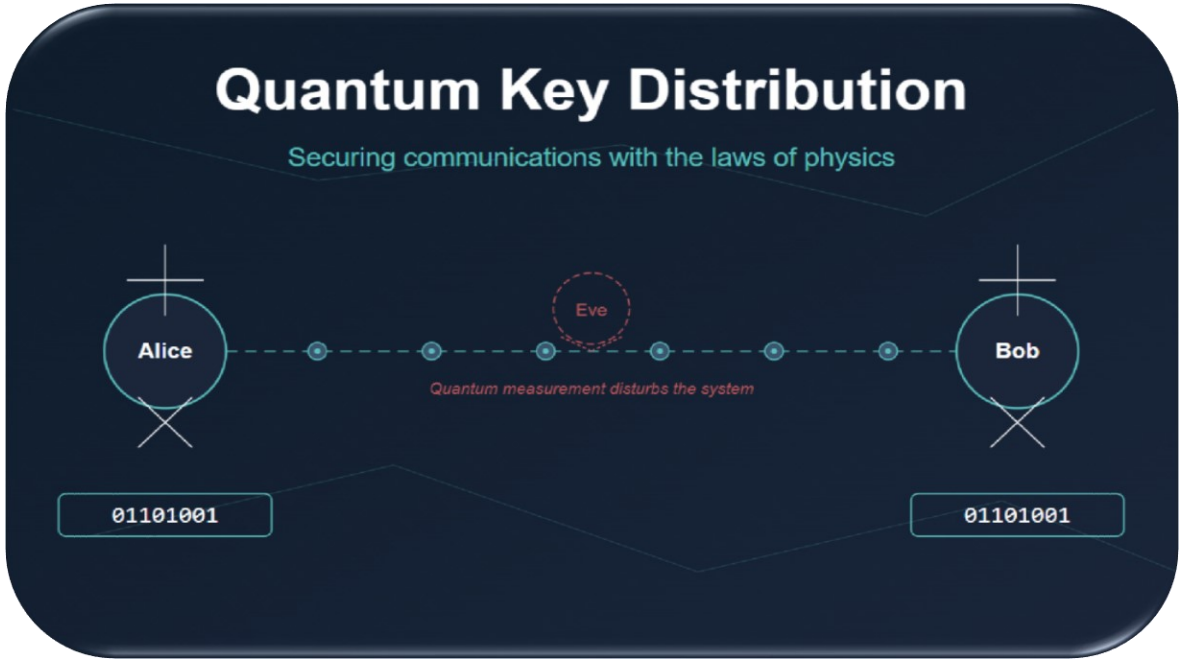


illustration Pranav Sanghadia (April 2025)

(tested by Geneva State for electronic vote between 2007 and 2014, and for encrypted communication in 2026)

Current limitations and challenges

Quantum computing is still in its NISQ (Noisy Intermediate-Scale Quantum) phase. The main challenges are :

- ❑ Improved stability (coherence time), reliability, and number of qubits
- ❑ More affordable error correction methods (cost and infrastructure)
- ❑ Discovery of more useful quantum algorithms

Progress is rapid, but it will take years (or decades depending on whom you ask) to develop fault-tolerant quantum computers capable of solving problems beyond the reach of classical computers.

Actions undertaken by Geneva State

Actions undertaken :

- ❑ First tests on quantum computers of IBM
- ❑ Participation of OCSIN to the Geneva Quantum Network
- ❑ Creation of a group of stakeholders at Geneva State dedicated to quantum initiatives

Actions in progress :

- ❑ Securing long-term access to quantum computers
- ❑ Defining suitable use-cases for Geneva State

Discussions with various partners :

REPUBLIQUE ET CANTON DE GENEVE
POST TENEBRAS LUX

hepia
Haute école du paysage, d'ingénierie et d'architecture de Genève

IBM Quantum

IDQ
an IonQ company

CERN

ETH zürich

EPFL

gesda

D-Wave
The Quantum Computing Company™

Thank you !

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